

HOJUN (DAN) YOO

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EDUCATION

Purdue University

Ph.D. in Computer Science (Advisor: Prof. Jason Wu)

West Lafayette, IN

Aug 2026 – Present

University of Notre Dame

B.S. in Computer Science, cum laude (GPA: 3.894 / 4.0)

Notre Dame, IN

Aug 2022 – May 2026

- **Tau Beta Pi**, National Engineering Honor Society

RESEARCH INTERESTS

Human–AI and LLM reasoning: (1) human–AI interaction, value alignment, and evaluation in HCI; (2) AI agents, tool use, and evaluation of interactive decision-making; (3) LLM reasoning, controllable behavior, and interpretability.

RESEARCH EXPERIENCE

SaNDwich Lab (IBM–Notre Dame collaboration), University of Notre Dame

Notre Dame, IN

Undergraduate Researcher (Advisor: Prof. Toby Jia-Jun Li)

Aug 2025 – May 2026

- Built a reproducible web-agent benchmark across retail, maps, and flight tasks with standardized GUI APIs, step logs, value personas, and causal ablations across interface and recommender conditions; co-authored an **IUI 2026** paper.
- Created trajectory-level evaluation and analysis tools (similarity, t-tests/ANOVA with effect sizes, robust regression, and deterministic harnesses), showing that salient UI cues can override values and deliberation predicts alignment better than task success.
- Developed a visual analytics system for evaluation by persona, model, and task using trace clustering, interactive filtering, and rationale/outcome visualization; extended manuscript in preparation for **CHI 2027**.

MINE Lab, University of Notre Dame

Notre Dame, IN

Undergraduate Researcher (Advisor: Prof. Xiangliang Zhang)

Sep 2024 – May 2026

- Co-first authored an **AAAI 2027** manuscript on sentence-level hidden-state interventions that increase reflection while preserving other behaviors; built semantic-aware directions using question-only baselines, orthogonal subtraction, stability filtering, and sentence-scope control.
- Released a reproducible toolkit (vector extraction, stability filtering, per-layer ablations) and authored ML primers and a hands-on tutorial for the NSF Cybertraining for Chemical Data Scientists project (Award #2321054).

FORESEER Lab, University of Michigan

Remote

Undergraduate Researcher (Advisor: Prof. Qiaozhu Mei)

May 2025 – May 2026

- Harmonized *LiveBench*, *HELM*, and *LMSYS Arena* by standardizing model IDs and deduplicating variants, then constructed model-similarity maps with MDS, centrality, and fit metrics.
- Diagnosed cross-benchmark misalignment, verified structural stability under noise and sub-sampling, and proposed compact task subsets that preserve similarity while reducing evaluation cost.

Verifier Development Group (alpha-beta-CROWN), UIUC

Urbana–Champaign, IL

Undergraduate Researcher (Advisor: Prof. Huan Zhang)

May 2025 – Aug 2025

- Built a batch runner and log parser for reproducible $\alpha\beta$ -CROWN robustness experiments, and streamlined model-loading and custom-operator demos into a one-command setup.
- Tracked down and patched a bug that was breaking L2 robustness checks, providing an immediate workaround and documenting a longer-term fix for the team.

Software Testing and Analysis group, UIUC

Remote, United States

Undergraduate Researcher (Advisor: Prof. Darko Marinov)

Jun 2024 – Sep 2024

- Automated build, test, and change injection across 9 OSS projects and analyzed 79 mutations plus 39 manually reviewed cases to map error propagation and inform targeted CI test selection.
- Curated a standardized mutation dataset (99 labeled cases; 52 cleaned/improved), giving the community a reusable benchmark that lowers replication cost and enables apples-to-apples comparison of future testing techniques.

TEACHING EXPERIENCE

Purdue University

Teaching Assistant, CS593: Graduate Human-Computer Interaction

West Lafayette, IN

Aug 2026 – Present

PUBLICATIONS AND WORKING PAPERS

Gebreegziabher, S. A., Yang, Y., Chiang, C., **Yoo, H.**, Chen, C., Do, H. J., Ashktorab, Z., Geyer, W., Gómez-Zarà, D., Li, T. J.-J. *The Behavioral Fabric of LLM-Powered GUI Agents: Human Values and Interaction Outcomes*. In **Proceedings of IUI 2026**, pp. 909–927 (co-author).

Yang, Y., Gebreegziabher, S. A., **Yoo, H.**, Chiang, C., Chen, C., Szymanski, A., Do, H. J., Ashktorab, Z., Geyer, W., Gómez-Zarà, D., Li, T. J.-J. *EvalAgent: Interactive Comparative Evaluation of Computer-Using GUI Agents*. In **Companion Proceedings of IUI 2026**, pp. 171–175 (Demos Track); extended manuscript in preparation for **CHI 2027** (co-author).

Yoo, H.*, Zhuang, H.* *Meaning-Removed Steering Vectors for Calibrating LLM Reasoning*. In preparation for **AAAI 2027** (co-first author).

* denotes equal contribution.

WORK EXPERIENCE

Move University

Seoul, South Korea

Independent Contractor (IC)

May 2026 – Aug 2026

- Drove end-to-end delivery of AI-powered customer acquisition and support workflows for Move University and Netpax, turning customer conversations into validated quotes, service requests, and ERP-ready records.
- Unified fragmented pricing and operating rules into auditable, versioned quote engines with deterministic calculations, RAG guidance, and safeguards against invalid pricing, unsupported responses, and data leakage.
- Productionized the platforms across six codebases with identity verification, payments, media evidence, admin tools, analytics, and automated end-to-end release gates.

FORGED

Notre Dame, IN

Co-founder / CTO

Jun 2025 – May 2026

- Built a normalized nutrition/supplement data platform with one-brand-one JSON records, NoSQL loaders, and OCR/LLM label parsing that mapped ingredients to safety tags and powered price, protein-cost, and macro comparisons.
- Added reliability and guardrails with schema validation, CI, structured logs, idempotent re-runs, zero-downtime index updates, and evidence-based listing policies so the catalog is trustworthy by default.

Tweeds

Remote

Software Engineering Intern

Nov 2024 – May 2025

- Built a book-publishing creator backend with secure, indexed MongoDB credentials and TikTok/Instagram APIs; added asynchronous jobs, retries, unified logs, and error handling for resilient ingest and promotion workflows.
- Improved QA and traceability with an end-to-end test harness (covering credential validation, upload, and polling), reusable user-seeding scripts, and centralized API logging (Winston) for the AI-detection pipeline.

Quantum AI

Seoul, South Korea

Software Engineering Intern

May 2023 – Jul 2023

- Scoped and prototyped a voice-support MVP, containerizing Asterisk PBX in Docker and integrating a Python speech-to-text/text-to-speech pipeline with a standardized Q&A knowledge base.
- Hardened for scale with a multithreaded call handler and load balancer, and shipped an admin console for agent management and analytics to improve reliability and operational visibility.

Republic of Korea Army

Goseong, Gangwon, South Korea

Sergeant, Reconnaissance Troop (55th Brigade, 22nd Infantry Division)

Jul 2020 – Jan 2022

- Served as a squad leader at a DMZ guard post, developing leadership and cross-functional teamwork under pressure.

LEADERSHIP, SERVICE AND OTHERS

O'Neill Family Hall, University of Notre Dame

Notre Dame, IN

Coding Tutor for Students

Aug 2022 – Dec 2024

- Provided one-on-one and small group tutoring in programming courses, helping peers strengthen coding fundamentals.

CS for Good, University of Notre Dame

Notre Dame, IN

Team Leader – Faith in Indiana Team / The Family Connection Team

Jan 2022 – May 2023

- Collected business-fair data with Python and automated event setup using Eventbrite/Zapier/Canva.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript (Node.js), Java, Bash, SQL

ML/NLP: PyTorch, HuggingFace Transformers, ONNX, NumPy, Pandas, scikit-learn

Eval/Stats & Viz: SciPy, statsmodels, NetworkX, Matplotlib, Seaborn, Pytest

Web/Data: Flask, React, SQLAlchemy, MongoDB, REST APIs, Asterisk (PBX)